

Chapter 4

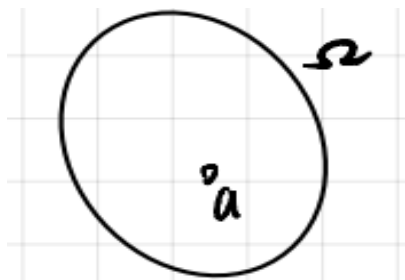
Cauchy Integral

1. Cauchy Integral Theorem

If $f(z)$ is analytic, then $\oint f(z) dz = 0$

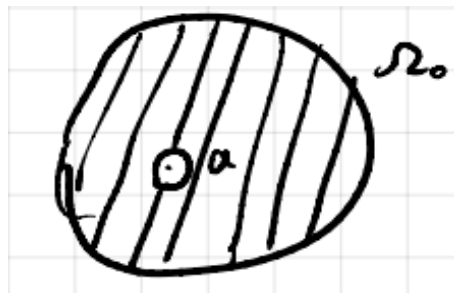
$$g(z) = \frac{f(z)}{z-a}, a \in \Omega$$

$$\oint g(z) dz = \oint \frac{f(z)}{z-a} dz = 2\pi i f(a)$$



proof

Make a small circle inside Ω with radius ε



$\frac{f(z)}{z-a}$ is analytic over Ω_0 . Then according to **Cauchy Theorem**, we have

$$\oint_{C_0} \frac{f(z)}{z-a} dz = 0 = \oint_C \frac{f(z)}{z-a} dz - \oint_{|z-a|<\varepsilon} \frac{f(z)}{z-a} dz$$

$$z = a + \varepsilon e^{i\theta}$$

$$dz = \varepsilon e^{i\theta} \cdot i d\theta$$

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$$\frac{dz}{d\theta} = \varepsilon e^{i\theta} \cdot i$$

For $\varepsilon \rightarrow 0$

$$\oint \frac{f(z)}{z-a} dz = \int_0^{2\pi} \frac{f(z)}{\varepsilon e^{i\theta}} \varepsilon e^{i\theta} \cdot i d\theta = 2\pi i f(a)$$

Then,

$$f(b) = \frac{1}{2\pi i} \oint \frac{f(z)}{z-b} dz$$

□

$$f(a) = \frac{1}{2\pi i} \oint \frac{f(z)}{z-a} dz$$

The function value on the boundary is depend on the region inside.

Example 1 find $\oint_{|z|=2} \frac{1+e^{\sin z^2}}{z-1} dz$

$$f(z) = 1 + e^{\sin z^2}$$

which is analytic, then

$$\oint_{|z|=2} \frac{f(z)}{z-1} dz = 2\pi i (1 + e^{\sin 1})$$

Example 2 find $\oint_{|z-2|=1} \frac{\sin(z^2+4)}{z-2.5} dz$

$$\begin{aligned} \oint_{|z-2|=1} \frac{\sin(z^2+4)}{z-2.5} dz &= 2\pi i f(a) \\ &= \sin(z^2+4)|_{z=2.5} \cdot 2\pi i \\ &= \sin(2.5^2+4) \cdot 2\pi i \end{aligned}$$

Example 3 find $\oint_{|z-2|=1} \frac{\sin(z^2+4)}{z-4} dz$

The point is outside the region, so $f(z)$ is analytic, then according to **Cauchy Theorem**, we have

$$\oint_{|z-2|=1} \frac{\sin(z^2 + 4)}{z - 4} dz = 0$$

Example 4

(1) find $\oint_{|z|=2} \frac{\cos z}{z} dz$

$$\oint_{|z|=2} \frac{\cos z}{z} dz = 2\pi i \cos(0) = 2\pi i$$

(2) find $\oint_{|z-1|=0.5} \frac{\sin(\pi z)}{(z-1)(z+1)} dz$

$$\oint_{|z-1|=0.5} \frac{\sin(\pi z)}{(z-1)(z+1)} dz = 2\pi i \frac{\sin(\pi a)}{a+1} \Big|_{a=1} = 0$$